

Problem 1.1.1

PROMPT

Given the tensors

$$\mathbf{A} = \begin{pmatrix} 1 & 2 & 0 \\ 2 & 4 & 1 \\ 0 & 1 & 3 \end{pmatrix} \text{ and } \mathbf{b} = \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix}$$

Solve:

SOLUTION

a) $\mathbf{A}_{ii}$

$$\mathbf{A}_{ii} \rightarrow \sum_i^3 \mathbf{A}_{ii} = A_{11} + A_{22} + A_{33} = \text{trace}(\mathbf{A}) = 8$$

b) $\mathbf{A}_{ij}\mathbf{b}_j$

$$\begin{aligned} c_i = A_{ij}b_j &\rightarrow \sum_j^3 A_{ij}b_j = A_{i1}b_1 + A_{i2}b_2 + A_{i3}b_3 = \\ &= \begin{pmatrix} A_{11}b_1 + A_{12}b_2 + A_{13}b_3 \\ A_{21}b_1 + A_{22}b_2 + A_{23}b_3 \\ A_{31}b_1 + A_{32}b_2 + A_{33}b_3 \end{pmatrix} = \begin{pmatrix} 1 \\ 4 \\ 6 \end{pmatrix} \end{aligned}$$

c) $\mathbf{A}_{ji}\mathbf{b}_j$

$$\begin{aligned} c_i = A_{ji}b_j &\rightarrow \sum_j^3 A_{ji}b_j = A_{1i}b_1 + A_{2i}b_2 + A_{3i}b_3 = \\ &= \begin{pmatrix} A_{11}b_1 + A_{21}b_2 + A_{31}b_3 \\ A_{12}b_1 + A_{22}b_2 + A_{32}b_3 \\ A_{13}b_1 + A_{23}b_2 + A_{33}b_3 \end{pmatrix} = \begin{pmatrix} 1 \\ 4 \\ 6 \end{pmatrix} \end{aligned}$$

Note: results for (b) and (c) are the same by coincidence

d) $\mathbf{b}_i\mathbf{b}_j$

$$A_{ij} = b_i b_j = \begin{pmatrix} b_1 b_1 & b_1 b_2 & b_1 b_3 \\ b_2 b_1 & b_2 b_2 & b_2 b_3 \\ b_3 b_1 & b_3 b_2 & b_3 b_3 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 2 \\ 0 & 0 & 0 \\ 2 & 0 & 4 \end{pmatrix}$$

e) $\mathbf{A}_{ij}\mathbf{A}_{ji}$

$$\alpha = \sum_i^3 \sum_j^3 A_{ij}A_{ji} =$$

$$\begin{aligned}
& A_{11}A_{11} + A_{12}A_{21} + A_{13}A_{31} + \\
& A_{21}A_{12} + A_{22}A_{22} + A_{23}A_{32} + \\
& A_{31}A_{13} + A_{32}A_{23} + A_{33}A_{33} \\
& = 36
\end{aligned}$$

Problem 1.1.2

PROMPT

Given the expressions in index notation, write the corresponding expressions in abstract tensor notation.

SOLUTION

a) $a_i b_i$

$$\alpha = a_i b_i = \sum_i^3 a_i b_i = \mathbf{a} \cdot \mathbf{b}$$

b) $a_i b_k c_k$

$$c_i = a_i \sum_k^3 b_k c_k \rightarrow \mathbf{c} = \mathbf{a} \cdot (\mathbf{b} \cdot \mathbf{c})$$

c) $A_{ik} B_{kj}$

$$C_{ij} = A_{ik} B_{kj} =$$

$$\sum_k^3 A_{ik} B_{kj} \rightarrow$$

$$\begin{pmatrix} A_{11}B_{11} + A_{12}B_{21} + A_{13}B_{31} & A_{11}B_{12} + A_{12}B_{22} + A_{13}B_{32} & A_{11}B_{13} + A_{12}B_{23} + A_{13}B_{33} \\ A_{21}B_{11} + A_{22}B_{21} + A_{23}B_{31} & A_{21}B_{12} + A_{22}B_{22} + A_{23}B_{32} & A_{21}B_{13} + A_{22}B_{23} + A_{23}B_{33} \\ A_{31}B_{11} + A_{32}B_{21} + A_{33}B_{31} & A_{31}B_{12} + A_{32}B_{22} + A_{33}B_{32} & A_{31}B_{13} + A_{32}B_{23} + A_{33}B_{33} \end{pmatrix}$$

$$= \mathbf{A} \cdot \mathbf{B}$$

d) $A_{ki} B_{kl} A_{lj}$

Assuming the answer as D_{ij} , since only i and j are non repeating, assume

$$C_{il} = A_{ki} B_{kl} \rightarrow \mathbf{C} = \mathbf{A}^T \cdot \mathbf{B}$$

Then

$$D_{ij} = C_{il} A_{lj} \rightarrow \mathbf{D} = \mathbf{C} \cdot \mathbf{A}$$

Finally

$$\mathbf{D} = \mathbf{A}^T \cdot \mathbf{B} \cdot \mathbf{A}$$

Another approach is:

Assume

$$E_{kj} = B_{kl}A_{lj} \rightarrow E = B \cdot A$$

Then

$$D_{ij} = A_{ki}E_{kj} \rightarrow D = A^T \cdot E$$

Which, of course, results in the same value!

$$D = A^T \cdot B \cdot A$$

e) $a_i A_{jk} b_k$

Let the answer be a matrix D_{ij} .

Let $c_j = A_{jk} b_k$

$$c_j = A_{j1}b_1 + A_{j2}b_2 + A_{j3}b_3 \rightarrow c = A \cdot b$$

then,

$$D_{ij} = a_i c_j \rightarrow \begin{pmatrix} a_1 c_1 & a_1 c_2 & a_1 c_3 \\ a_2 c_1 & a_2 c_2 & a_2 c_3 \\ a_3 c_1 & a_3 c_2 & a_3 c_3 \end{pmatrix} = a \cdot c^T$$

Thus

$$D = a \cdot (A \cdot b)^T$$

f) $a_i A_{ik} b_k$

From the previous question, we know that

$$c_i = A_{ik} b_k \rightarrow c = A \cdot b$$

And, simply

$$\alpha = a_i c_i = a \cdot c$$

The answer is

$$\alpha = a \cdot A \cdot b$$

Problem 1.2.1

PROMPT

Show that $\delta_{ij}\delta_{ji} = 3$.

SOLUTION

$$\delta_{ij}\delta_{ji} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \cdot \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}^T = 1 \cdot 1 + 1 \cdot 1 + 1 \cdot 1 = 3$$

Problem 1.2.2

PROMPT

Find the expression in index notation for

$$\frac{\partial \mathbf{v}}{\partial \mathbf{v}}$$

where $\mathbf{v}$ is a vector.

SOLUTION

$$\mathbf{A} = \frac{\partial \mathbf{v}}{\partial \mathbf{v}} = \begin{pmatrix} \frac{\partial v_1}{\partial v_1} & \frac{\partial v_1}{\partial v_2} & \frac{\partial v_1}{\partial v_3} \\ \frac{\partial v_2}{\partial v_1} & \frac{\partial v_2}{\partial v_2} & \frac{\partial v_2}{\partial v_3} \\ \frac{\partial v_3}{\partial v_1} & \frac{\partial v_3}{\partial v_2} & \frac{\partial v_3}{\partial v_3} \end{pmatrix} \rightarrow A_{ij} = \frac{\partial v_i}{\partial v_j}$$

It is known that

$$\frac{\partial v_l}{\partial v_m} = \begin{cases} 1 & \text{if } l = m \\ 0 & \text{else} \end{cases}$$

Then it is safe to assume that

$$\frac{\partial v_i}{\partial v_j} = \delta_{ij}$$

Problem 1.2.3

PROMPT

With the aid of the Kronecker delta, find the expression in index notation for the trace of $\mathbf{a}\mathbf{b}^T$ where $\mathbf{a}$ and $\mathbf{b}$ are vectors.

SOLUTION

$$\alpha = \text{trace}(\mathbf{a}\mathbf{b}^T) = a_i b_j \delta_{ij}$$

With the contraction property

$$b_j \delta_{ij} = b_i$$

$$\alpha = a_i \cdot b_i = \mathbf{a} \cdot \mathbf{b}$$

Problem 1.2.4

PROMPT

Simplify the expression

$$\frac{\partial^2}{\partial x_i \partial x_j} (x_i x_j)$$

SOLUTION

$$\begin{aligned} \frac{\partial^2}{\partial x_i \partial x_j} (x_i x_j) &= \frac{\partial}{\partial x_i} \left(\frac{\partial x_i}{\partial x_j} x_j + \frac{\partial x_j}{\partial x_j} x_i \right) = \\ &= \frac{\partial}{\partial x_i} (0 + \delta_{jj} x_i) = 3 \frac{\partial}{\partial x_i} (x_i) = 3 \delta_{ii} = 9 \end{aligned}$$

Problem 1.3.1

PROMPT

Show that

$$\epsilon_{ijk} a_i a_j = 0$$

SOLUTION

By brute force...

Let $A_{ij} = a_i a_j$

$$\left(\begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 & -1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \right) \cdot \begin{pmatrix} a_1 a_1 & a_1 a_2 & a_1 a_3 \\ a_2 a_1 & a_2 a_2 & a_2 a_3 \\ a_3 a_1 & a_3 a_2 & a_3 a_3 \end{pmatrix} = \begin{pmatrix} +a_2 a_3 - a_3 a_2 \\ -a_1 a_3 + a_3 a_1 \\ +a_1 a_2 - a_2 a_1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

By triviality

Multiplying both sides by the cartesian basis vector:

$$\epsilon_{ijk} a_i a_j \cdot \hat{e}_i = 0 \cdot \hat{e}_i = 0$$

The expression above is the cross product of the vector $\mathbf{a}$.

$$\epsilon_{ijk} a_i a_j \cdot \hat{e}_i = \mathbf{a} \times \mathbf{a}$$

By definition, the cross product of a vector by itself is 0, since they are colinear.

Problem 1.3.2

PROMPT

Prove de identity

$$\epsilon_{ijk}\epsilon_{mnk} = \delta_{im}\delta_{jn} - \delta_{in}\delta_{jm}$$

SOLUTION

$$\epsilon_{ijk}\epsilon_{mnk}$$

The product of two Levi-Civita tensors can be written as the determinant of a 3×3 matrix of Kronecker deltas:

$$\epsilon_{ijk}\epsilon_{mnp} = \det \begin{pmatrix} \delta_{im} & \delta_{in} & \delta_{ip} \\ \delta_{jm} & \delta_{jn} & \delta_{jp} \\ \delta_{km} & \delta_{kn} & \delta_{kp} \end{pmatrix}$$

We aim to solve the specific case where $p = k$:

$$\epsilon_{ijk}\epsilon_{mnk} = \det \begin{pmatrix} \delta_{im} & \delta_{in} & \delta_{ik} \\ \delta_{jm} & \delta_{jn} & \delta_{jk} \\ \delta_{km} & \delta_{kn} & \delta_{kk} \end{pmatrix}$$

Expanding the determinant along the bottom row, and knowing that $\delta_{kk} = 3$ and $\delta_{ik}\delta_{km} = \delta_{im}$, the matrix simplifies directly to the identity:

$$\begin{aligned} \epsilon_{ijk}\epsilon_{mnk} &= \det \begin{pmatrix} \delta_{im} & \delta_{in} \\ \delta_{jm} & \delta_{jn} \end{pmatrix} \\ &= \delta_{im}\delta_{jn} - \delta_{in}\delta_{jm} \end{aligned}$$

Problem 1.3.3

PROMPT

Using the permutation operator, prove the vector triple product identity

$$\mathbf{a} \times (\mathbf{b} \times \mathbf{c}) = \mathbf{b}(\mathbf{a} \cdot \mathbf{c}) - \mathbf{c}(\mathbf{a} \cdot \mathbf{b})$$

SOLUTION

Let $\mathbf{r}$ be the result vector.

$$\mathbf{r} = \mathbf{a} \times (\mathbf{b} \times \mathbf{c}) \rightarrow r_i = \epsilon_{ijk}a_jd_k$$

For $d_k = \epsilon_{kmn}b_m c_n$

Therefore

$$r_i = \epsilon_{ijk} a_j \epsilon_{kmn} b_m c_n$$

From the property obtained from the last problem, and the fact that $\epsilon_{kmn} = \epsilon_{mnk}$, we know that:

$$\begin{aligned}\epsilon_{ijk} \epsilon_{mnk} &= \delta_{im} \delta_{jn} - \delta_{in} \delta_{jm} \\ r_i &= (\delta_{im} \delta_{jn} - \delta_{in} \delta_{jm}) a_j b_m c_n \\ &= a_j b_m c_n \delta_{im} \delta_{jn} - a_j b_m c_n \delta_{in} \delta_{jm} \\ &= b_i (a_j c_j) - c_i (a_j b_j)\end{aligned}$$

Translating the scalar components back to vector notation, where $(a_j c_j) = (\mathbf{a} \cdot \mathbf{c})$:

$$\mathbf{r} = \mathbf{b}(\mathbf{a} \cdot \mathbf{c}) - \mathbf{c}(\mathbf{a} \cdot \mathbf{b})$$

Problem 1.4.1

PROMPT

Given two vectors $\mathbf{a}$ and $\mathbf{b}$, show that

$$(\mathbf{a} \otimes \mathbf{a})(\mathbf{b} \otimes \mathbf{b}) = (\mathbf{a} \cdot \mathbf{b}) \mathbf{a} \otimes \mathbf{b}$$

SOLUTION

Let the tensor $\mathbf{R}$ be the result.

$$\begin{aligned}\mathbf{R} &= (\mathbf{a} \otimes \mathbf{a})(\mathbf{b} \otimes \mathbf{b}) \rightarrow R_{ij} = a_i a_k b_k b_j \\ &= (a_k b_k) a_i b_j \\ R_{ij} &= (a_k b_k) a_i b_j \rightarrow \mathbf{R} = (\mathbf{a} \cdot \mathbf{b}) \cdot \mathbf{a} \otimes \mathbf{b}\end{aligned}$$

Problem 1.4.2

PROMPT

Find the abstract expression for

$$\frac{\partial}{\partial \mathbf{v}} (\mathbf{v} \cdot \mathbf{v})$$

SOLUTION

$$\begin{aligned}
\frac{\partial}{\partial \mathbf{v}}(\mathbf{v} \cdot \mathbf{v}) &\rightarrow \frac{\partial}{\partial v_j}(v_i v_i) = 2 \frac{\partial v_i}{\partial v_j} v_i \\
&= 2 \delta_{ij} v_i \\
&= 2 v_j
\end{aligned}$$

Therefore

$$\frac{\partial}{\partial \mathbf{v}}(\mathbf{v} \cdot \mathbf{v}) = 2\mathbf{v}$$

Problem 1.4.3

PROMPT

Given two second-order tensor $\mathbf{A}$ and $\mathbf{B}$, show that

$$\frac{\partial}{\partial \mathbf{A}}(\mathbf{A} : \mathbf{B}) = \mathbf{B}$$

SOLUTION

In index notation:

$$\begin{aligned}
\frac{\partial}{\partial \mathbf{A}}(\mathbf{A} : \mathbf{B}) &\rightarrow \frac{\partial}{\partial A_{ij}}(A_{mn} B_{mn}) \\
\frac{\partial}{\partial A_{ij}}(A_{mn} B_{mn}) &= \frac{\partial A_{mn}}{\partial A_{ij}} B_{mn} + \frac{\partial B_{mn}}{\partial A_{ij}} A_{mn} \\
&= \delta_{mi} \delta_{nj} B_{mn} \\
&= B_{ij} \\
&\rightarrow \mathbf{B}
\end{aligned}$$

Problem 1.4.4

PROMPT

Find the expression for

$$\frac{\partial}{\partial \mathbf{a}}(\mathbf{a} \times \mathbf{b})$$

SOLUTION

In index notation,

$$\begin{aligned}\frac{\partial}{\partial \mathbf{a}}(\mathbf{a} \times \mathbf{b}) &\rightarrow \frac{\partial}{\partial a_l}(\epsilon_{ijk} a_j b_k) = \epsilon_{ijk} b_k \delta_{jl} \\ &= \epsilon_{ilk} b_k = -\epsilon_{ikl} b_k\end{aligned}$$

Problem 1.4.5

PROMPT

Compute the derivative, with respect to $\mathbf{A}$, of

$$\mathbf{A} - \frac{1}{3} \text{trace}(\mathbf{A}) \mathbf{I}$$

SOLUTION

$$\begin{aligned}\frac{1}{3} \frac{\partial}{\partial A_{mn}} (A_{ij} - A_{kk} \delta_{ij}) &= \delta_{im} \delta_{jn} - \frac{1}{3} \delta_{ij} \delta_{km} \delta_{kn} \\ &= \delta_{im} \delta_{jn} - \frac{1}{3} \delta_{ij} \delta_{mn} \\ &= \mathbb{I}_{ijmn} - \frac{1}{3} \delta_{ij} \delta_{mn} \\ &\rightarrow \mathbb{I} - \frac{1}{3} \mathbf{I} \otimes \mathbf{I}\end{aligned}$$

Problem 1.5.1

PROMPT

Using index notation, show that

SOLUTION

a) $\mathbf{I} : \mathbf{I} = 3$

Since $\mathbf{A} : \mathbf{B} \rightarrow A_{ij} B_{ij}$, then our solution is:

$$\begin{aligned}\alpha &= \delta_{ij} \delta_{ij} \\ &= \delta_{ii} \\ &= 3\end{aligned}$$

b) $\mathbb{I} : \mathbb{I} = \mathbb{I}$

By definition, $\mathbb{I}_{ijkl} = \delta_{ik} \delta_{jl}$. Thus

$$\begin{aligned}
\mathbb{I}_{ijkl}\mathbb{I}_{klmn} &= (\delta_{ik}\delta_{jl})(\delta_{km}\delta_{ln}) \\
&= \delta_{im}\delta_{jn} \\
&= \mathbb{I}
\end{aligned}$$

c) $\mathbb{I}^{\text{sym}} : \mathbb{I}^{\text{sym}} = \mathbb{I}^{\text{sym}}$

$$\begin{aligned}
\mathbb{I}_{ijmn}^{\text{sym}}\mathbb{I}_{mnkl}^{\text{sym}} &= \frac{1}{2}(\delta_{im}\delta_{jn} + \delta_{in}\delta_{jm}) \cdot \frac{1}{2}(\delta_{mk}\delta_{nl} + \delta_{ml}\delta_{nk}) \\
&= \frac{1}{4}(\delta_{im}\delta_{jn}\delta_{mk}\delta_{nl} + \delta_{im}\delta_{jn}\delta_{ml}\delta_{nk} + \delta_{in}\delta_{jm}\delta_{mk}\delta_{nl} + \delta_{in}\delta_{jm}\delta_{ml}\delta_{nk}) \\
&= \frac{1}{4}(\delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk} + \delta_{il}\delta_{jk} + \delta_{ik}\delta_{jl}) \\
&= \frac{1}{2}(\delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk}) \\
&= \mathbb{I}_{ijkl}^{\text{sym}}
\end{aligned}$$